

NOAH SIEGEL

GitHub: <https://github.com/nasiegel88>

siegelanoah@gmail.com

Phone: +1 707 418 5077

LinkedIn: <https://www.linkedin.com/in/noah-siegel>

EDUCATION

University of California, Davis, Davis, CA

Sept 18 - Nov 23

PhD, Immunology; Minor: Bioinformatics

November, 2023

Advisor: Lisa Miller (PI)

GPA: 3.7

California State University, East Bay, CA

Aug 15 - May 18

BS, Biology; Minor: Chemistry

June, 2018

GPA: 3.7

TECHNICAL STRENGTHS

Computer Languages: R, Bash, Python

Software & Tools: Snakemake, R Shiny, AWS, Slurm, Docker, Cursor, Git, LaTeX, Microsoft applications

Operating systems: Windows, Linux

Lab bench: DNA/RNA extraction, PCR, NGS Library prep

EXPERIENCE

First Tracks Biotherapeutics (formally AnaptysBio)

Apr 24 - Present

- Developed computational pipelines to analyze genetic and proteomic data
- Delivered bioinformatics support to project leads, enabling data-driven decision making
- Deployed and maintained AWS infrastructure for hosting internal and publicly available data analyses
- Developed tools to determine mechanisms of action for internal biologics
- Developed tools to guide indication selection for in-licensed biologics

General Dynamics Information Technology

Dec 23 - Jan 24

- Contributed to large-scale data analysis projects
- Assessed pathogen virulence factors from human samples
- Developed whole genome sequencing pipelines

UC Davis

Apr 19 - Dec 23

- Led epidemiological studies in rhesus monkeys
- Developed NGS workflows for transcriptomic analysis
- Conducted microbiome analysis in the rhesus monkey model
- Characterized the metabolic profile of the infant rhesus monkey gut microbiome using metabolomics

Tilden Prep

Jun 18 - Sept 18

- Taught introductory physics to high school students and led them through laboratory practicals

PEER REVIEWED PUBLICATIONS

1. NA Siegel, MT Jimenz, CS Rocha, M Rolston, S Dandekar, JV Solnick, and LA Miller. *Helicobacter pylori* infection in infant rhesus macaque monkeys is associated with an altered lung and oral microbiome. *Scientific Reports* 2024.

MANUSCRIPTS IN REVIEW/PENDING SUBMISSION

1. NA Siegel, KA Thongphanh, LA Miller. Effects of Environmental Change on Airway Sensitization: Recent Insights and Future Perspectives. *J Asthma and Allergy* (year TBD)
2. NA Siegel, T Westmont, M Rolston, S Dandekar, H Deshmukh, LA Miller. Early-life antibiotic treatment results in persistent microbiome alterations in association with sex. *mSphere* (year TBD).
3. NA Siegel, DT Dugger, V Kohli, H Ji, MD Pena-Ponce, J Ngo, and LA Miller. Transcriptome Analysis of Postnatal Airway Epithelium Reveals a Gene Signature of Differentiation and Host Defense that is Distinct from Adult Airway Epithelium (journal TBD, year TBD).

HONORS AND AWARDS

P51 Diversity Fellowship

RELEVANT COURSES

Core Courses

Advanced Immunology
Mucosal Immunology
Host-microbe Interactions

Other Courses

Bioinformatics
Biostatistics
Genomics
Single Cell RNA-Seq Analysis
Metagenomics and Metatranscriptomics

PRESENTATIONS AT SCIENTIFIC MEETINGS

1. Noah A. Siegel, Matt Ralston, Taylor Westmont, Alexa Rindy, Hitesh Deshmukh, Lisa A. Miller. *Effect of Neonatal Antibiotic Treatment on the Respiratory Microbiome: Altered Development and Sexual Dimorphism*. Presented at the 2022 American Thoracic Society international conference in San Francisco
2. Noah A. Siegel, Matt Ralston, Taylor Westmont, Alexa Rindy, Hitesh Deshmukh, Lisa A. Miller. *Effect of Neonatal Antibiotic Treatment on the Respiratory Microbiome: Altered Development and Sexual Dimorphism*. Presented at the 2022 Bay Area Microbial Pathogenesis Symposium
3. Noah A. Siegel, Matt Ralston, Taylor Westmont, Alexa Rindy, Hitesh Deshmukh, Lisa A. Miller. *Early Life Antibiotic Treatment Results in Both Transient and Persistent Changes to the Microbiome*. Presented at the 2021 American Thoracic Society international conference (virtual).
4. Noah A. Siegel, Matt Ralston, Taylor Westmont, Alexa Rindy, Hitesh Deshmukh, Lisa A. Miller. *Early Life Microbiome Perturbation Results in Persistent Composition and Functional Changes in Association with Sex*. Presented at the 2021 National Institute of Environmental Health Sciences Microbiome workshop (virtual).
5. Noah A. Siegel, Taylor Westmont, Hong Ji, Lisa A. Miller. *Muc5b Expression in Infant Airway Epithelium is Developmentally Regulated*. Presented at the 2020 University of California, Davis Lung Day conference (virtual).

6. Noah A. Siegel, Taylor Westmont, Daniel Dugger, Hong Ji, Vikram Kohli, and Lisa A. Miller. *Developmental Regulation of the MUC5B Mucin Sheds Light on Infant Susceptibility to Infection and Reduced Host-defense During Early Life*. Presented at the 2020 American Thoracic Society international conference (virtual).

UNIVERSITY SERVICE

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| 1. Positivity And Wellness for Students (PAWS) Committee | Oct 21 - Aug 23 |
| 2. Graduate Group in Immunology Executive Committee | Jun 20 - Jun 21 |
| 3. Graduate Group in Immunology Admissions Committee | Jun 19 - Jun 21 |

ADDITIONAL ACTIVITIES

1. Mentor-mentee coordinator, ESTEME
2. Volunteered at Holy Rosary Catholic School